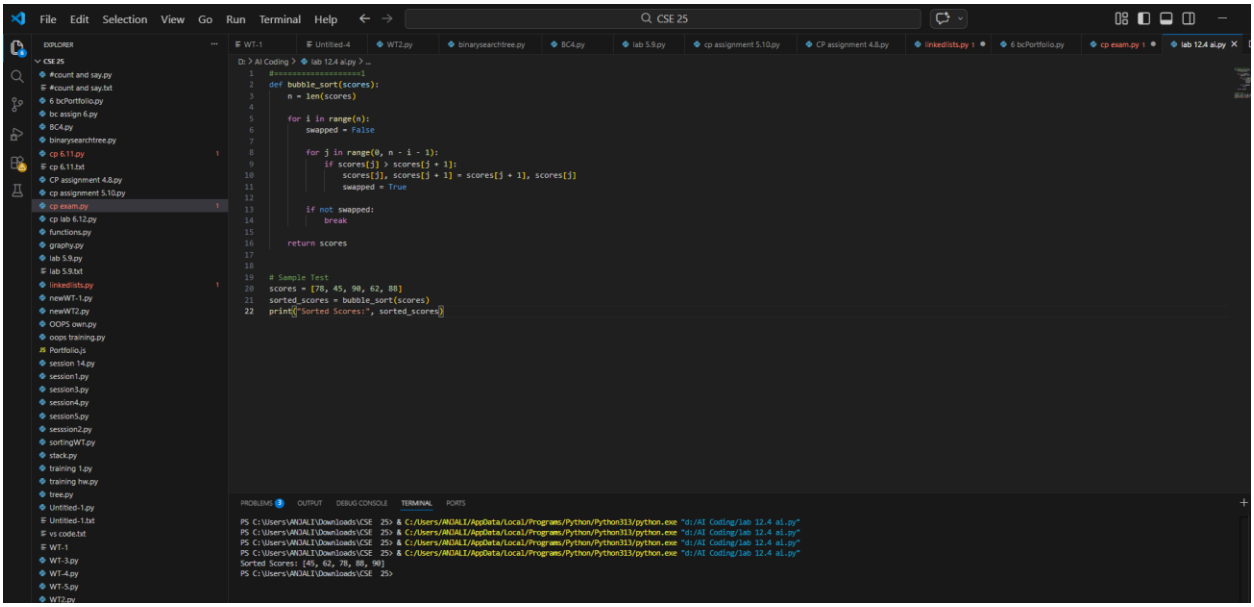
2303a51924

Lab assignment- 12.4

Task 1: Bubble Sort for Ranking Exam Scores

Scenario

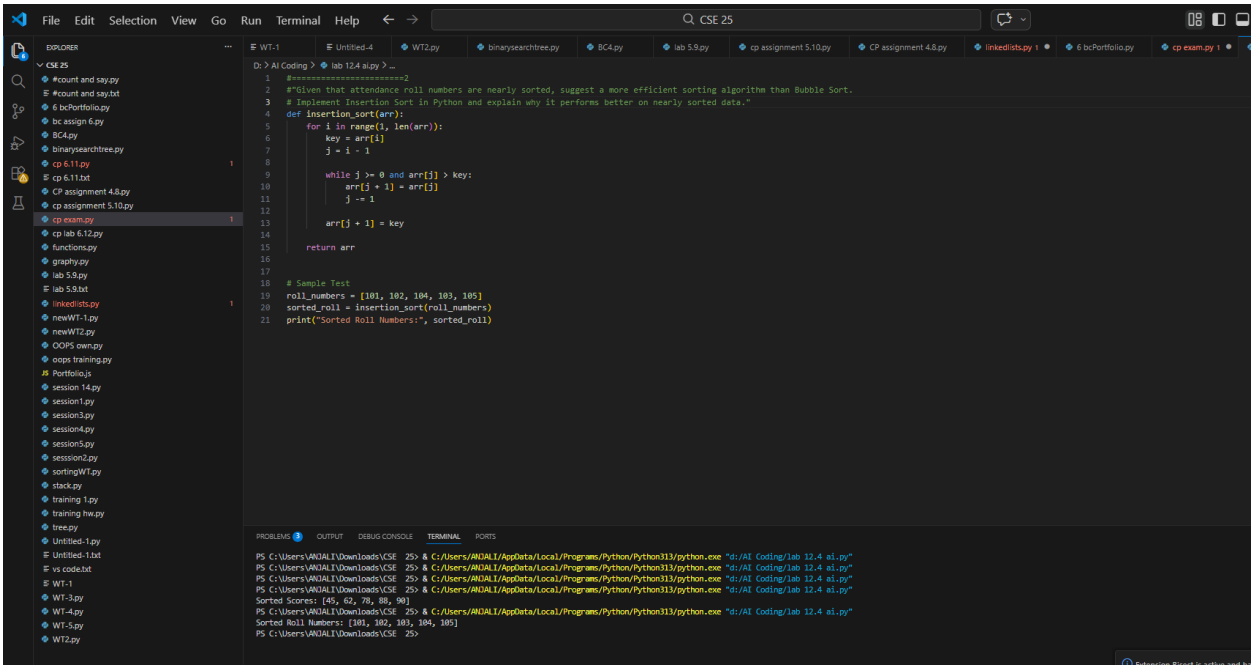
You are working on a college result processing system where a small list of student scores needs to be sorted after every internal assessment.



Task 2: Improving Sorting for Nearly Sorted Attendance Records

Scenario

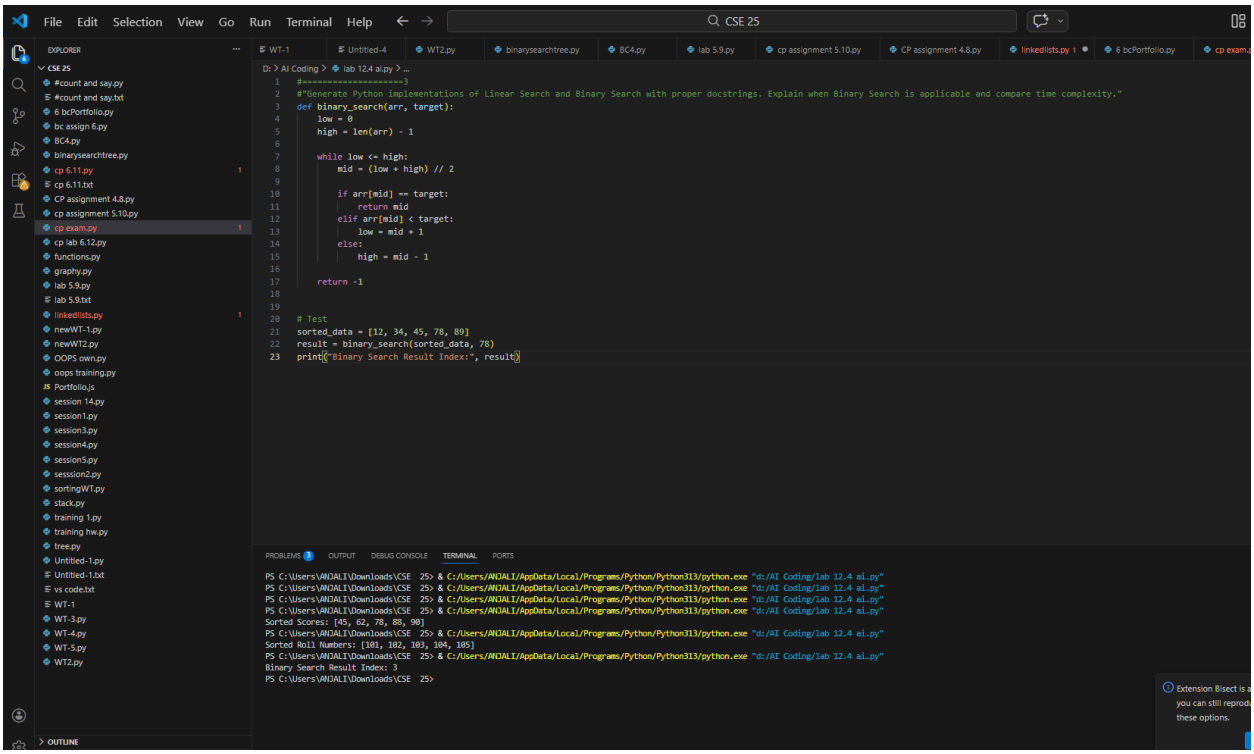
You are maintaining an attendance system where student roll numbers are already almost sorted, with only a few late updates.



Task 3: Searching Student Records in a Database

Scenario

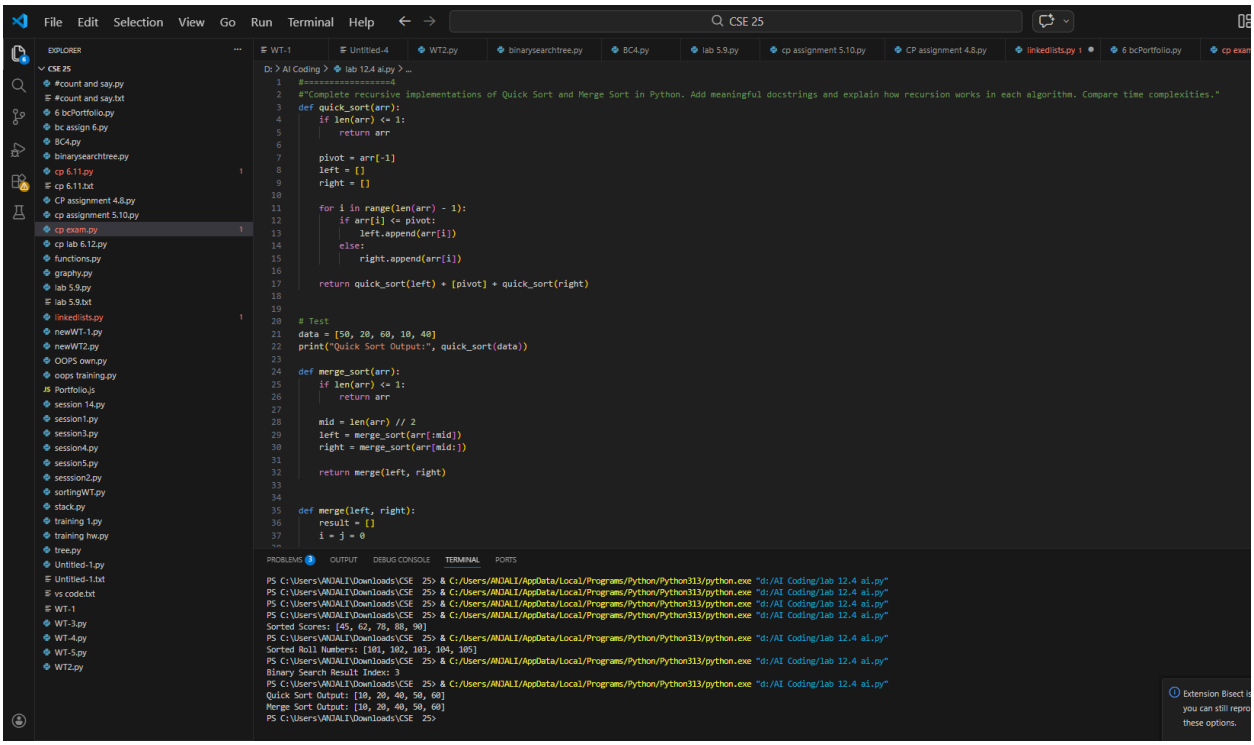
You are developing a student information portal where users search for student records by roll number.

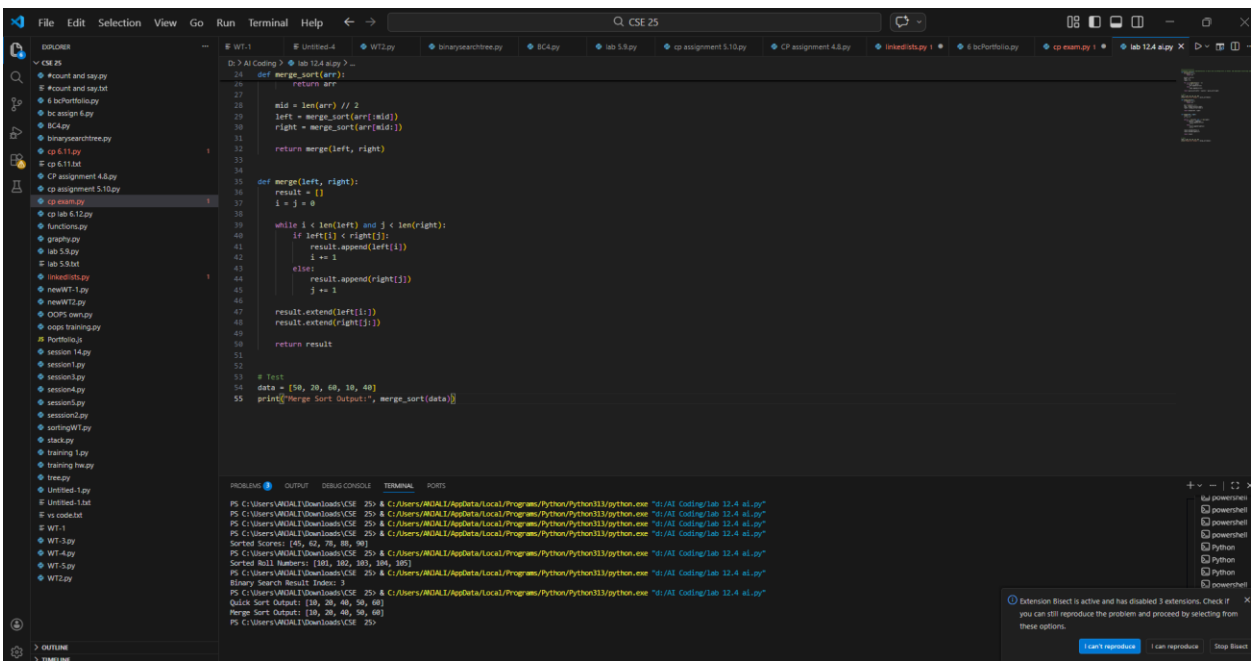


Task 4: Choosing Between Quick Sort and Merge Sort for Data Processing

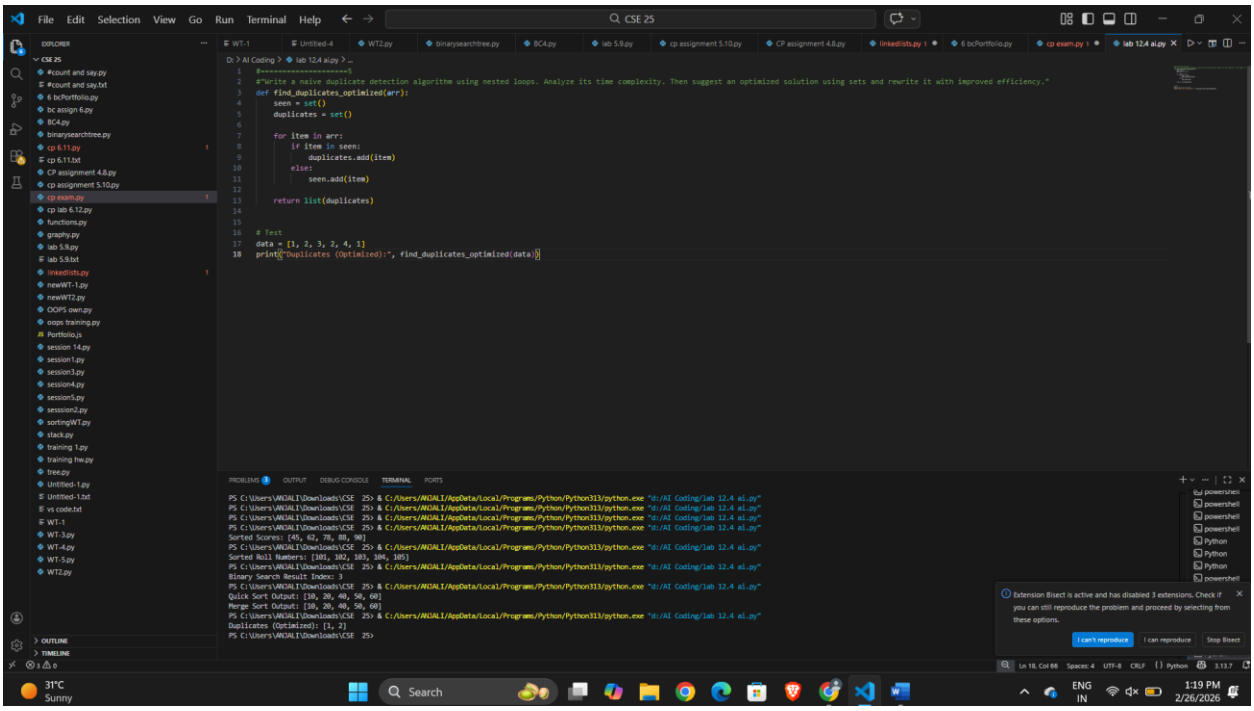
Scenario

You are part of a data analytics team that needs to sort large datasets received from different sources (random order, already sorted, and reverse sorted)





Task 5: Optimizing a Duplicate Detection Algorithm

Scenario

You are building a data validation module that must detect duplicate user IDs in a large dataset before importing it into a system.